ConQuest: Quiz Game System
Program Manual

Group Information

Project Name: ConQuest - Quiz Game System
Section: 4-305
Group Number: 5
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Project Description

ConQuest is an interactive Java-based Quiz Game System designed to test user's knowledge across multiple categories. The program provides a graphical user interface (GUI) that allows users to navigate through menus, answer randomized questions, and receive immediate feedback.

The system supports multiple quiz categories including General Knowledge, Brainrot (internet culture), Driving, Geography, Programming, and Media. It incorporates multimedia features such as images and audio to enhance user engagement.

Key features include:

- Randomized question selection
- Immediate feedback on answers
- Score calculation and result display
- Persistent score storage
- Interactive and user-friendly GUI navigation
- Audio and visual enhancements

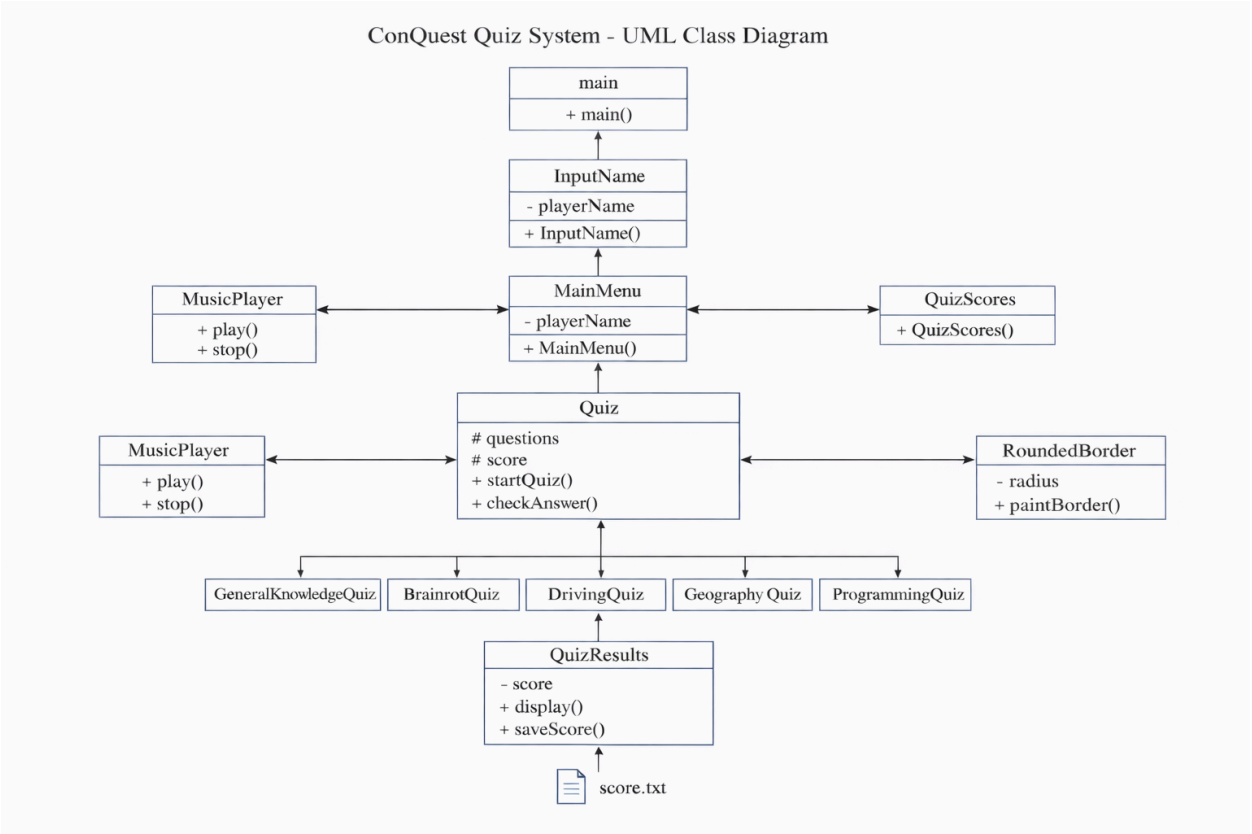
Program Structure

The system adopts a modular, class-based architecture to ensure organization, maintainability, and scalability. Program execution begins in the main class, which initializes the InputName interface to collect user input.

Navigation is handled through the MainMenu and QuizMenu classes, which guide the user through the application and provide access to the available quiz categories. Each quiz category is implemented as a separate class (e.g., GeneralKnowledgeQuiz, DrivingQuiz, BrainrotQuiz, etc.), allowing independent operation while maintaining a consistent structure for question handling, answer validation, and score computation.

Upon completion of a quiz, results are managed by the QuizResults class, which displays the user's performance and saves the data to a text file (score.txt). Previously saved results can be accessed through the QuizScores class, which presents the data in a structured table format.

Supporting classes such as MusicPlayer and RoundedBorder provide additional functionality. The MusicPlayer class handles background music and sound effects, while the RoundedBorder class is used to enhance the visual appearance of the user interface.

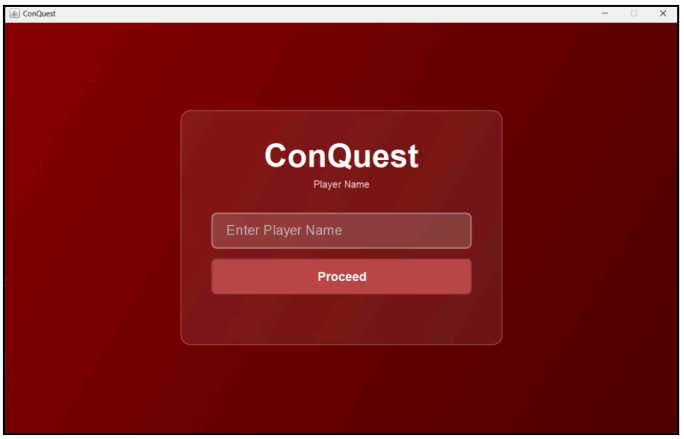


User Guide

This section provides step-by-step instructions for operating the ConQuest Quiz Game System, from launching the program to viewing saved results.

1. **Launching the Application**
   a. Open the project in a Java-supported IDE or run the compiled .jar file.
   b. Execute the main class.
   c. The application will display the Player Name Input Screen.

Figure 1: Player Name Input Screen



2. **Entering Player Name**
   a. Click the input field labeled “Enter Player Name”.

b. Type your desired player name.
c. Click “Proceed” or press “Enter”.

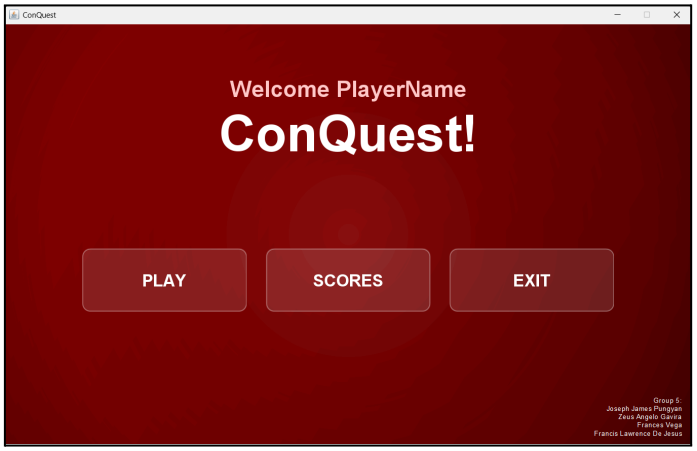
Notes:

- The field cannot be left blank.
- The default placeholder text is not accepted as valid input.

3. Navigating the Main Menu

After entering a valid name, the Main Menu is displayed.

Figure 2: Main Menu Interface



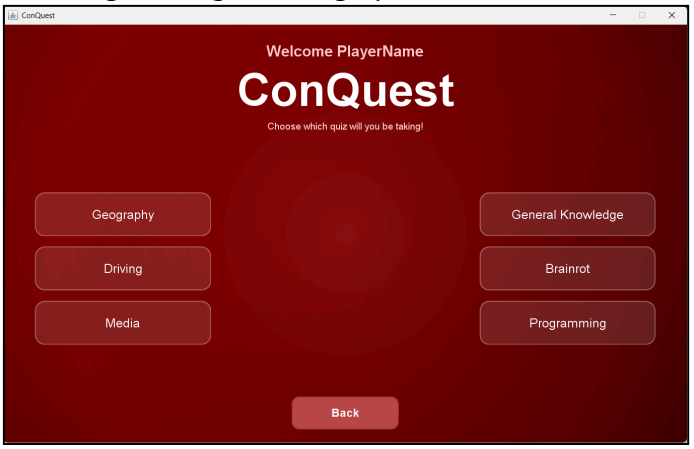
Available options:

- Play - Start a new quiz
- Scores - View previously saved results
- Exit - Close the application

4. Selecting a Quiz Category

a. Click “Play”.
b. The Quiz Menu appears with available categories.
c. Select one of the following:
   i. General Knowledge
   ii. Brainrot
   iii. Driving
   iv. Geography
   v. Programming
   vi. Media

Figure 3: Quiz Category Selection Screen

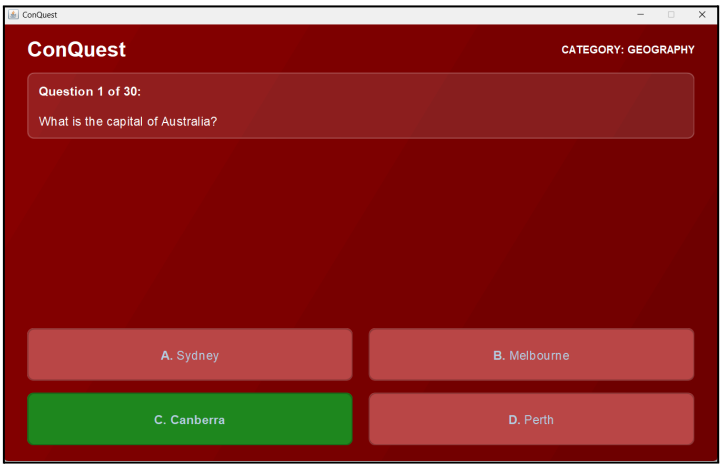


d. Confirm your selection to begin the quiz.

5. Taking the Quiz

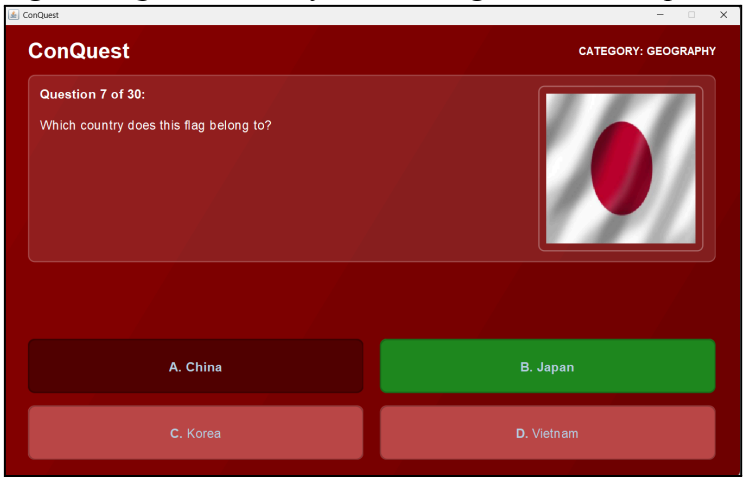
a. Each quiz session presents 30 randomized multiple-choice questions.
b. Answering questions:
   i. Read the question carefully.
   ii. Select one of the four choices (A-D).
   iii. Once selected, the answer is locked.
c. Correct answer feedback:
   i. The selected option is highlighted in green.

Figure 4: Question Interface Showing Correct Response



d. Incorrect answer feedback:
   i. The selected option is highlighted in red.
   ii. The correct answer is also revealed.

Figure 5: Question Interface Showing Incorrect Response

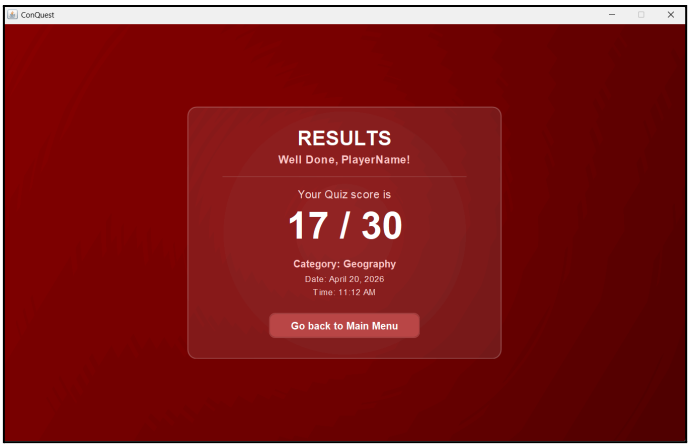


e. Progression:
   i. The quiz automatically proceeds to the next question after each response.
   ii. No manual navigation is required.

6. Viewing Quiz Results

a. After completing all questions, the Results Screen is displayed.

Figure 6: Quiz Results Display



b. Information shown:
   i. Player name
   ii. Final score (e.g., 17/30)
   iii. Quiz category
   iv. Date and time of completion
c. A sound effect is played to indicate completion.

7. Saving Results

a. Quiz results are automatically saved to:
   i. If run through the compiled .jar file: Conquest/score.txt.
   ii. If main.java run through Java-supported IDE: Conquest/ConquestUI/ConquestPkg/score.txt
b. Each record includes:

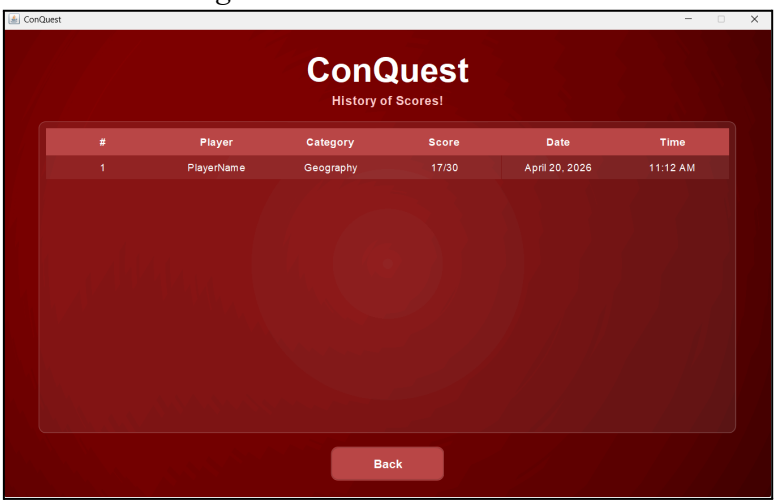
i. Player name
ii. Category
iii. Score
iv. Date
v. Time

c. No manual saving is required.

8. Viewing Score History

a. From the Main Menu, click Scores.
b. A table of previously recorded results will be displayed.

Figure 7: Saved Score Records



c. Notes:
   i. If no records exist, the message "No results Yet" will be shown.
   ii. Click "Back" to return to the Main Menu.

9. Exiting the Application

a. From the Main Menu, click "Exit".
b. Confirm the exit prompt to close the application.

Conclusion

The ConQuest Quiz Game System demonstrates a well-structured and fully functional application that integrates core programming concepts with an interactive graphical user interface. Through its modular design, the system effectively separates responsibilities across multiple classes, allowing for organized development, maintainability, and scalability.

The application successfully implements key features of a quiz system, including randomized question delivery, real-time answer validation, score computation, and persistent storage of results. The inclusion of multimedia elements such as images and audio further enhances user engagement, providing a more dynamic and intuitive experience.

Overall, ConQuest delivers a cohesive and user-friendly platform for knowledge assessment across multiple domains. Its design reflects both technical correctness and thoughtful user experience, making it a reliable and extensible solution for quiz-based applications.